AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 4 · STUDENT LESSON

Configuration and secrets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Typed contracts and schemas. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of configuration and secrets. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **configuration** and **secret**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **configuration** and **secret**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Typed contracts and schemas. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **configuration** and **secret**s using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Configuration and secrets case. Separate observations, given facts and assumptions before choosing an explanation.

Configuration supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Configuration and secrets. Connect each labelled input, deciding step and result.

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

Configuration and secrets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Configuration and secrets. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Change maximum_words to 60.

> **Predicted result**
>
> **Prediction**: The configured answer limit becomes 60; the private token stays unchanged.

> **Why the result follows**
>
> Reason: An ordinary length setting does not require replacing the credential.

Words that unlock the idea

> Use the vocabulary of Configuration and secrets to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Secret	A value whose disclosure can grant privileged access
Configuration	Declared settings controlling operation
Redaction	Removing sensitive values from displayed or recorded output

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Configuration and secrets, predict a step, then compare with its explanation.

A startup script prints every environment variable while debugging a missing data version.

It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Worked explanation · Configuration and secrets

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Configuration and secrets

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

The facts we will use

Supplied facts

maximum_words=80 is public **configuration**. A separate protected store holds a placeholder credential.

AI ACADEMY / AI-6/4

Configuration and secrets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Load setting

Read the reviewed maximum_words value of 80.



2. Load credential

The runtime obtains the token through its protected path.



3. Log safely

Record startup success without printing the token.

CHANGE ONE THING

Change maximum_words to 60.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Load setting	Read the reviewed maximum_words value of 80.
2. Load credential	The runtime obtains the token through its protected path.
3. Log safely	Record startup success without printing the token.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Configuration controls behaviour. A credential grants access and needs different storage and handling.

What this example does not establish

Keep the stated limit

Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

Keep the stated limit

Read the labelled picture: Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Keep the stated limit

Change one thing: Change maximum_words to 60. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Configuration and secrets claim and explain a repair using the supplied evidence.

A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Configuration and secrets with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Configuration** and **secret**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Configuration** and **secret**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Configuration and secrets. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Configuration and secrets. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled **configuration** and **secret**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **configuration** and **secret**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Configuration and secrets case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Configuration and secrets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

> Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

> Server **secret** exists, but a browser error includes its full value. What failed and what record

should be produced instead?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Configuration and secrets, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

By the end, I can	How I will show it
Explain the mechanism and terminology of configuration and secrets	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Configuration and secrets without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **configuration** and **secret**s

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record startup success without printing the token.

1. Load credential

2. Log safely

3. Load setting

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Log safely Record startup success without printing the token. **Configuration** controls behaviour. A credential grants access and needs different storage and handling.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change maximum_words to 60. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The configured answer limit becomes 60; the private token stays unchanged.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Repair a missing building block

Review Typed contracts and schemas (ai-6/3). Goal: Explain the **mechanism** and terminology of typed contracts and schemas. Short practice: Compare A's count 4 with B's count "4".

Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Configuration and secrets graphic using its labels, deciding evidence and relevant limitation.

01. Configuration and secrets: the case



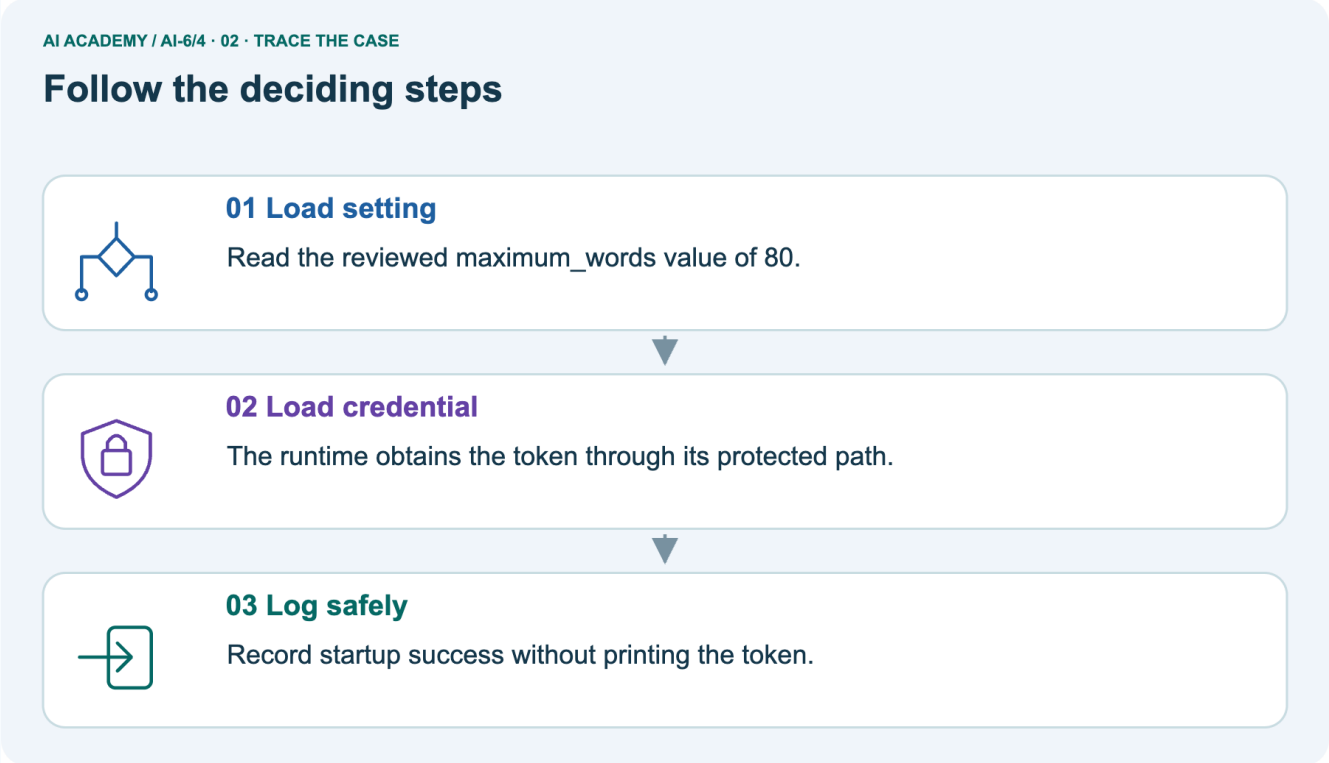
Follow the labelled picture with the matching explanation. Open the vector diagram

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Supplied facts: maximum_words=80 is public **configuration**.

A separate protected store holds a placeholder credential.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Load setting: Read the reviewed maximum_words value of 80. → Load credential: The runtime obtains the token through its protected path. → Log safely: Record startup success without printing the token.

Point to each numbered stage and explain which recorded **input** it uses.

03. Configuration and secrets — evidence trace

AI ACADEMY / AI-6/4 · 03 · INSPECT THE LESSON MODEL

Configuration and secrets — evidence trace

Input and scope	Check or calculation	Conclusion and limit
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration and **secret**s — **evidence** trace.

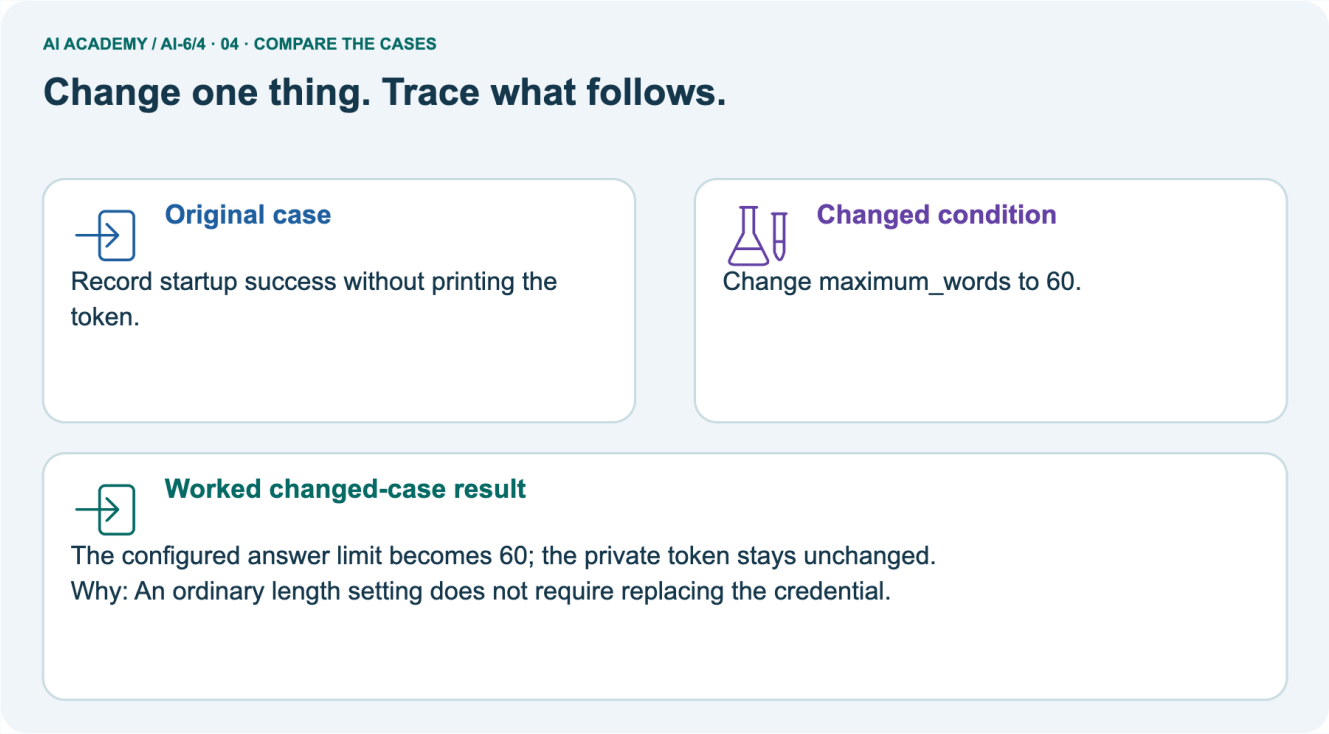
A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**.

Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**. | It risks exposing **secret**s unrelated to the error.

Report the missing key name without values and validate only required fields.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



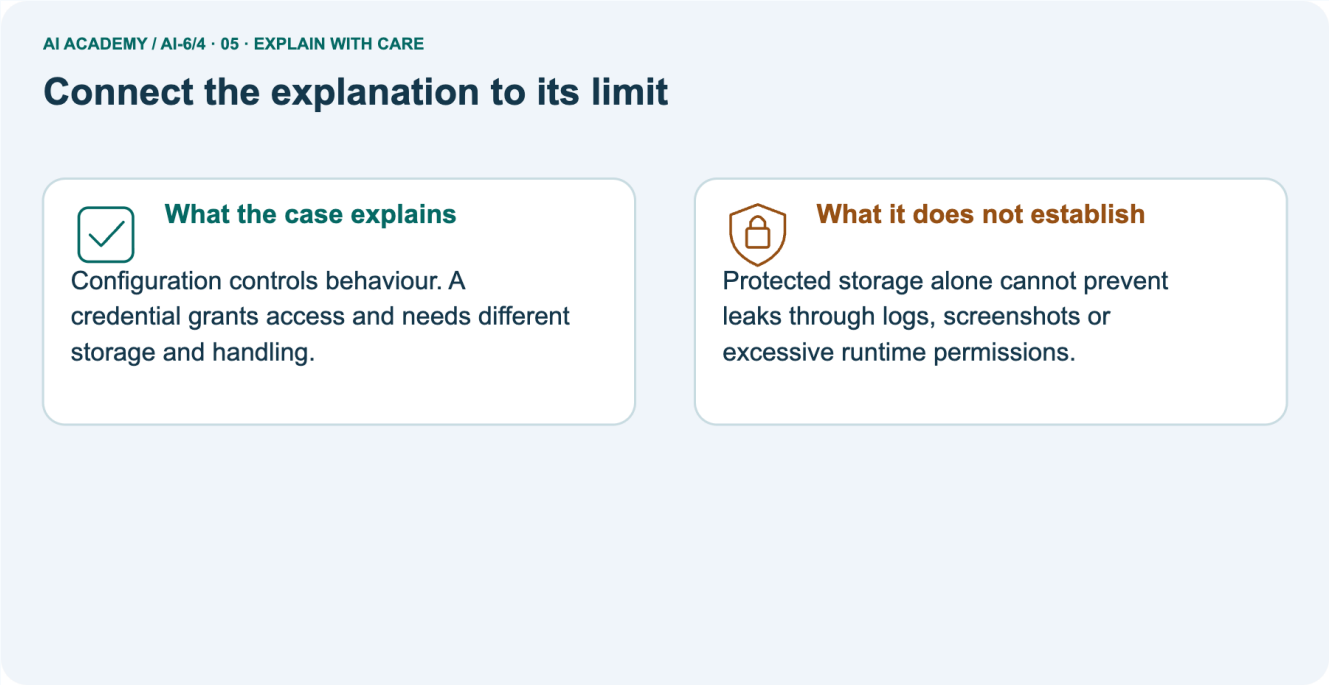
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record startup success without printing the token. Changed condition: Change maximum_words to 60. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

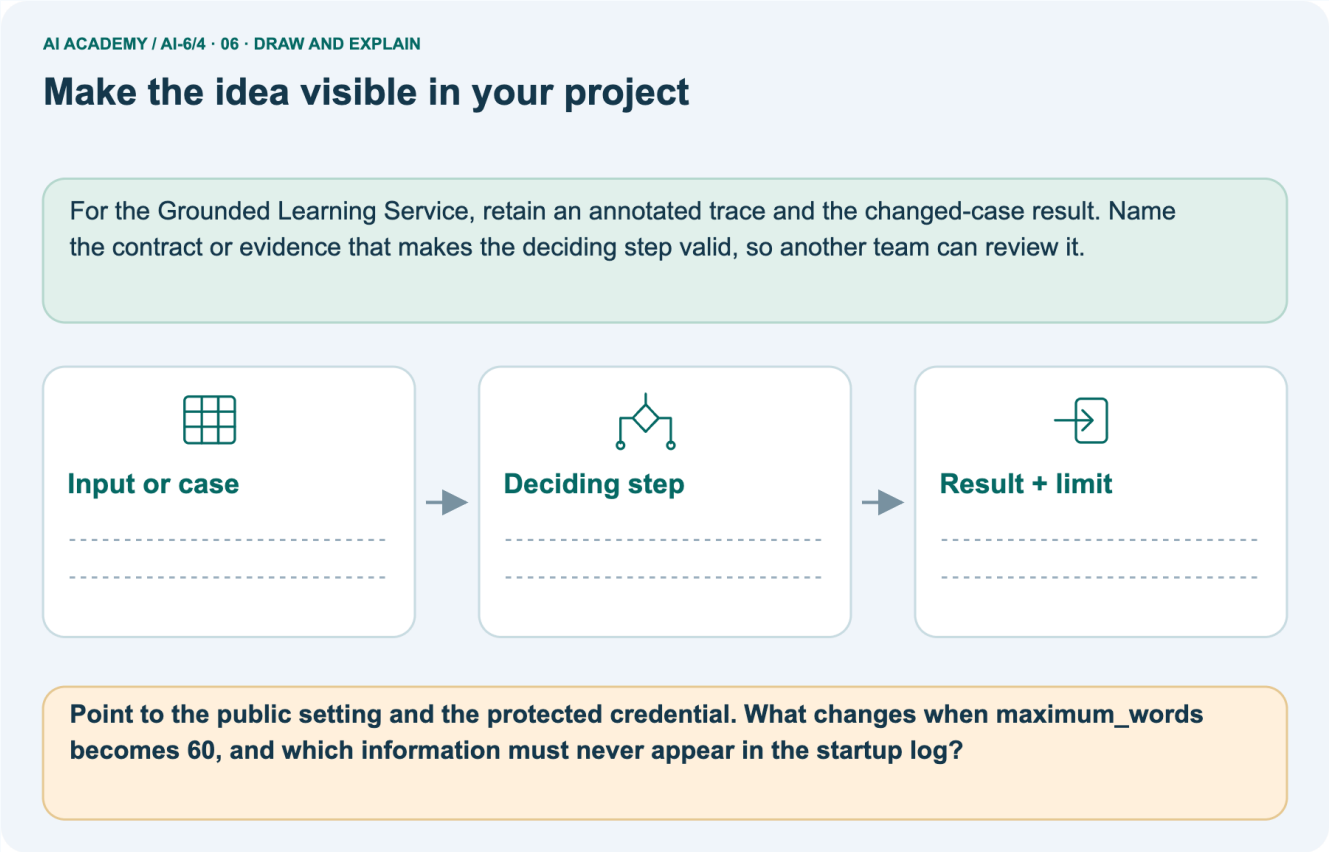


Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration controls behaviour. A credential grants access and needs different storage and handling. Limit: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.